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compact1, compact2, compact3

java.lang

Class RuntimePermission

java.lang.Object
  java.security.Permission
    java.security.BasicPermission
      java.lang.RuntimePermission

All Implemented Interfaces:

Serializable, Guard

```
public final class RuntimePermission
extends BasicPermission
```

This class is for runtime permissions. A RuntimePermission contains a name (also referred to as a "target name") but no actions list; you either have the named permission or you don't.

The target name is the name of the runtime permission (see below). The naming convention follows the hierarchical property naming convention. Also, an asterisk may appear at the end of the name, following a ".", or by itself, to signify a wildcard match. For example: "loadLibrary.*" and "*" signify a wildcard match, while "*loadLibrary" and "a*b" do not.

The following table lists all the possible RuntimePermission target names, and for each provides a description of what the permission allows and a discussion of the risks of granting code the permission.

Permission Target Name	What the Permission Allows	Risks of Allowing this Permission
createClassLoader	Creation of a class loader	This is an extremely dange permission to grant. Malic applications that can insta their own class loaders cou load their own rogue class

		the system. These newly lo classes could be placed int protection domain by the c loader, thereby automatica granting the classes the permissions for that domai
getClassLoader	Retrieval of a class loader (e.g., the class loader for the calling class)	This would grant an attack permission to get the class for a particular class. This dangerous because having to a class's class loader all attacker to load other class available to that class load attacker would typically ot not have access to those cl
setContextClassLoader	Setting of the context class loader used by a thread	The context class loader is system code and extension they need to lookup resour might not exist in the syste loader. Granting setContextClassLoader per would allow code to chang context class loader is use particular thread, includin threads.
enableContextClassLoaderOverride	Subclass implementation of the thread context class loader methods	The context class loader is system code and extension they need to lookup resour might not exist in the syste loader. Granting enableContextClassLoader permission would allow a s of Thread to override the m that are used to get or set context class loader for a particular thread.
closeClassLoader	Closing of a ClassLoader	Granting this permission a code to close any URLClas that it has a reference to.
setSecurityManager	Setting of the security manager (possibly replacing an existing one)	The security manager is a that allows applications to implement a security polic Granting the setSecurityM permission would allow co change which security ma used by installing a differe possibly less restrictive se manager, thereby bypassin

		that would have been enfo the original security mana
createSecurityManager	Creation of a new security manager	This gives code access to protected, sensitive metho may disclose information a other classes or the execut stack.
getenv.{variable name}	Reading of the value of the specified environment variable	This would allow code to re value, or determine the ex of a particular environmen variable. This is dangerous variable contains confiden
exitVM.{exit status}	Halting of the Java Virtual Machine with the specified exit status	This allows an attacker to denial-of-service attack by automatically forcing the v machine to halt. Note: The "exitVM.*" permission is automatically granted to a loaded from the applicatio path, thus enabling applica terminate themselves. Also "exitVM" permission is equ to "exitVM.*".
shutdownHooks	Registration and cancellation of virtual-machine shutdown hooks	This allows an attacker to a malicious shutdown hook interferes with the clean sh of the virtual machine.
setFactory	Setting of the socket factory used by ServerSocket or Socket, or of the stream handler factory used by URL	This allows code to set the implementation for the soc server socket, stream hand RMI socket factory. An atta may set a faulty implemen which mangles the data st
setIO	Setting of System.out, System.in, and System.err	This allows changing the v the standard system strea attacker may change Syste monitor and steal user inp may set System.err to a "n OutputStream, which woul any error messages sent to System.err.
modifyThread	Modification of threads, e.g., via calls to Thread `interrupt`, `stop`, `suspend`, `resume`, `setDaemon`, `setPriority`, `setName` and `setUncaughtExceptionHandler` methods	This allows an attacker to the behaviour of any threa system.

stopThread	Stopping of threads via calls to the Thread `stop` method	This allows code to stop ar in the system provided tha already granted permissio access that thread. This po threat, because that code r corrupt the system by killi existing threads.
modifyThreadGroup	modification of thread groups, e.g., via calls to ThreadGroup `destroy`, `getParent`, `resume`, `setDaemon`, `setMaxPriority`, `stop`, and `suspend` methods	This allows an attacker to thread groups and set thei priority.
getProtectionDomain	Retrieval of the ProtectionDomain for a class	This allows code to obtain information for a particula source. While obtaining po information does not comp the security of the system, give attackers additional information, such as local names for example, to bett an attack.
getFileSystemAttributes	Retrieval of file system attributes	This allows code to obtain system information such as usage or disk space availal the caller. This is potential dangerous because it discl information about the syst hardware configuration an information about the calle privilege to write files.
readFileDescriptor	Reading of file descriptors	This would allow code to re particular file associated w file descriptor read. This is dangerous if the file contai confidential data.
writeFileDescriptor	Writing to file descriptors	This allows code to write t particular file associated w descriptor. This is dangero because it may allow malic code to plant viruses or at least, fill up your entire dis
loadLibrary.{library name}	Dynamic linking of the specified library	It is dangerous to allow an permission to load native c libraries, because the Java architecture is not designe does not prevent malicious behavior at the level of nat code.

accessClassInPackage.{package name}	Access to the specified package via a class loader's `loadClass` method when that class loader calls the SecurityManager `checkPackageAccess` method	This gives code access to c packages to which it norm not have access. Malicious may use these classes to h attempt to compromise sec the system.
defineClassInPackage.{package name}	Definition of classes in the specified package, via a class loader's `defineClass` method when that class loader calls the SecurityManager `checkPackageDefinition` method.	This grants code permissio define a class in a particul package. This is dangerous because malicious code wi permission may define rog classes in trusted package `java.security` or `java.la` example.
accessDeclaredMembers	Access to the declared members of a class	This grants code permissio query a class for its public, protected, default (packag access, and private fields a methods. Although the cod have access to the private protected field and metho it would not have access to private/protected field data would not be able to invok private methods. Neverthe malicious code may use thi information to better aim a attack. Additionally, it may any public methods and/or public fields in the class. T be dangerous if the code w normally not be able to inv those methods and/or acce fields because it can't cast object to the class/interfac those methods and fields.
queuePrintJob	Initiation of a print job request	This could print sensitive information to a printer, or waste paper.
getStackTrace	Retrieval of the stack trace information of another thread.	This allows retrieval of the trace information of anoth thread. This might allow m code to monitor the execut threads and discover vulnerabilities in applicati
setDefaultUncaughtExceptionHandler	Setting the default handler to be used when a thread	This allows an attacker to a malicious uncaught exce

	terminates abruptly due to an uncaught exception	handler that could interfer termination of a thread
preferences	Represents the permission required to get access to the java.util.prefs.Preferences implementations user or system root which in turn allows retrieval or update operations within the Preferences persistent backing store.)	This permission allows the read from or write to the preferences backing store user running the code has sufficient OS privileges to write to that backing store actual backing store may r within a traditional filesyst directory or within a regist depending on the platform
usePolicy	Granting this permission disables the Java Plug-In's default security prompting behavior.	For more information, refe Plug-In's guides, Applet Se Basics and usePolicy Perm

See Also:

BasicPermission, Permission, Permissions, PermissionCollection, SecurityManager, Serialized Form

Constructor Summary

Constructors

Constructor	Description
`RuntimePermission(String name)`	Creates a new RuntimePermission with the specified name.
`RuntimePermission(String name, String actions)`	Creates a new RuntimePermission object with the specified name.

Method Summary

Methods inherited from class java.security.BasicPermission

equals, getActions, hashCode, implies, newPermissionCollection

Methods inherited from class java.security.Permission

checkGuard, getName, toString

Methods inherited from class java.lang.Object

clone, finalize, getClass, notify, notifyAll, wait, wait, wait

Constructor Detail

RuntimePermission

```
public RuntimePermission(String name)
```

Creates a new RuntimePermission with the specified name. The name is the symbolic name of the RuntimePermission, such as "exit", "setFactory", etc. An asterisk may appear at the end of the name, following a ".", or by itself, to signify a wildcard match.

Parameters:

name - the name of the RuntimePermission.

Throws:

NullPointerException - if name is null.

IllegalArgumentException - if name is empty.

RuntimePermission

```
public RuntimePermission(String name,
                         String actions)
```

Creates a new RuntimePermission object with the specified name. The name is the symbolic name of the RuntimePermission, and the actions String is currently unused and should be null.

Parameters:

name - the name of the RuntimePermission.

actions - should be null.

Throws:

NullPointerException - if name is null.

IllegalArgumentException - if name is empty.

Submit a bug or feature
For further API reference and developer documentation, see Java SE Documentation. That documentation contains more detailed, developer-targeted descriptions, with conceptual overviews, definitions of terms, workarounds, and working code examples.
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